GYROSCOPIC COUPLE AND PRECESSIONAL MOTION

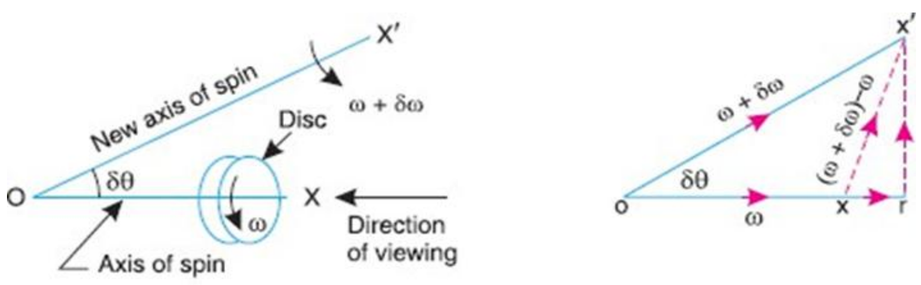
Introduction -

1. When a body moves along a curved path with a uniform linear velocity, a force in the direction of centripetal acceleration (known as centripetal force) must be applied externally over the body, so that it moves along the required curved path. This external force applied is known as active force.

2.When a body, itself, is moving with uniform linear velocity along a circular path, it is subjected to the centrifugal force* radially outwards. This centrifugal force is called reactive force. The action of the reactive or centrifugal force is to tilt or move the body along radially outward direction.

Precessional Angular Motion

The angular acceleration is the rate of change of angular velocity with respect to time. It is a vector quantity and may be represented by drawing a vector diagram with the help of right-hand screw rule.



(a) (b)

Consider a disc, as shown in Fig (a), revolving or spinning about the axis OX (known as axis of spin) in anticlockwise when seen from the front, with an angular velocity ω in a plane at right angles to the paper.

After a short interval of time δt, let the disc be spinning about the new axis of spin OX ′(at an angle δθ) with an angular velocity (ω+ δω). Using the right-hand screw rule, initial angular velocity of the disc (ω) is represented by vector ox; and the final angular velocity of the disc (ω+ δω) is represented by vector ox′ as shown in Fig. 14.1 (b). The vector xx′ represents the change of angular velocity in time δt i.e., the angular acceleration of the disc. This may be resolved into two components, one parallel to ox and the other perpendicular to ox

Component of angular acceleration in the direction of ox,

$$\alpha_t = \frac{xr}{\delta t} = \frac{or - ox}{\delta t} = \frac{ox' \cos \delta\theta - ox}{\delta t}$$

$$= \frac{(\omega + \delta\omega) \ \cos\delta\theta - \omega}{\delta t} = \frac{\omega\cos\delta\theta + \delta\omega \cos\delta\theta - \omega}{\delta t}$$

Since $\delta\theta$ is very small, therefore substituting $\cos \delta\theta = 1$, we have

$$\alpha_t = \frac{\omega + \delta\omega - \omega}{\delta t} = \frac{\delta\omega}{\delta t}$$

In the limit, when $\delta t \to 0$,

$$\alpha_t = \underset{\delta t \to 0}{\text{Lt}} \left(\frac{\delta\omega}{\delta t}\right) = \frac{d\omega}{dt}$$

Component of angular acceleration in the direction perpendicular to ox,

$$\alpha_c = \frac{rx'}{\delta t} = \frac{ox' \sin\delta\theta}{\delta t} = \frac{(\omega + \delta\omega)\sin\delta\theta}{\delta t} = \frac{\omega\sin\delta\theta + \delta\omega.\sin\delta\theta}{\delta t}$$

Since $\delta\theta$ in very small, therefore substituting $\sin\delta\theta = \delta\theta$, we have

$$\alpha_c = \frac{\omega.\delta\theta + \delta\omega.\delta\theta}{\delta t} = \frac{\omega.\delta\theta}{\delta t}$$

...(Neglecting $\delta\omega.\delta\theta$, being very small)

In the limit when $\delta t \to 0$,

$$\alpha_c = \underset{\delta t \to 0}{\text{Lt}} \frac{\omega.\delta\theta}{\delta t} = \omega \times \frac{d\theta}{dt} = \omega.\omega_P \qquad \ldots\left(\text{Substituting } \frac{d\theta}{dt} = \omega_P\right)$$

$\therefore$ Total angular acceleration of the disc

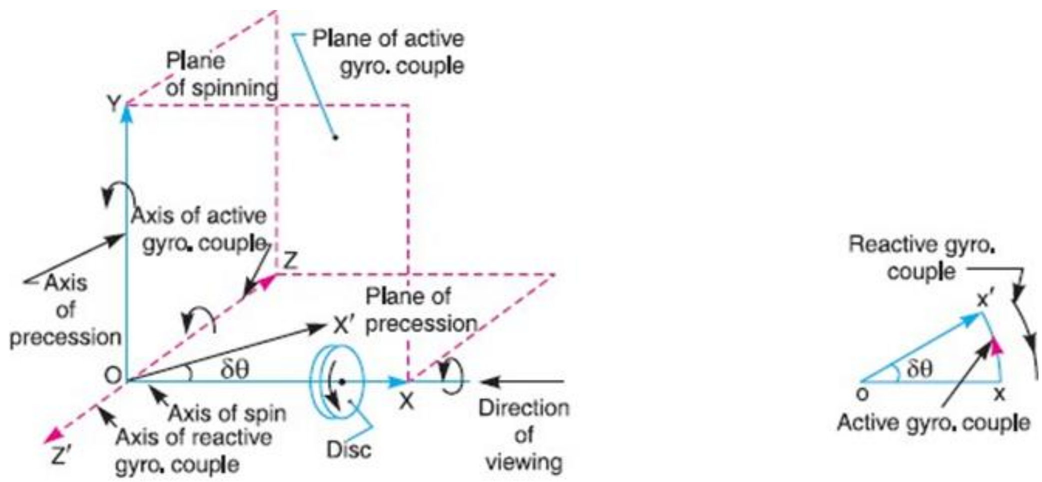
$$= \text{vector } xx' = \text{vector sum of } \alpha_t \text{ and } \alpha_c$$

$$= \frac{d\omega}{dt} + \omega \times \frac{d\theta}{dt} = \frac{d\omega}{dt} + \omega.\omega_P$$

where $d\theta/dt$ is the angular velocity of the axis of spin about a certain axis, which is perpendicular to the plane in which the axis of spin is going to rotate. This angular velocity of the axis of spin ($d\theta/dt$) is known as angular velocity of precession and is denoted by ωP. The axis, about which the axis of spin is to turn, is known as axis of precession. The angular motion of the axis of spin about the axis of precession is known as precessional angular motion.

Gyroscopic Couple -

Consider a disc spinning with an angular velocity ω rad/s about the axis of spin OX, in anticlockwise direction when seen from the front, as shown in Fig.(a). Since the plane in which the disc is rotating is parallel to the plane YOZ, therefore it is called plane of spinning. The plane XOZ is a horizontal plane, and the axis of spin rotates in a plane parallel to the horizontal plane about an axis OY. In other words, the axis of spin is said to be rotating or processing about an axis OY. In other words, the axis of spin is said to be rotating or processing about an axis OY (which is perpendicular to both the axes OX and OZ) at an angular velocity _P rap/s. This horizontal plane XOZ is called plane of precession and OY is the axis of precession.



(a) (b)

Let I = Mass moment of inertia of the disc about OX, and ω= Angular velocity of the disc.

Angular momentum of the disc = I. ω

Since the angular momentum is a vector quantity, therefore it may be represented by the vector ox′, as shown in Fig.(b). The axis of spin OX is also rotating anticlockwise when seen from the top about the axis OY. Let the axis OX is turned in the plane XOZ through a small angle δθ radians to the position OX′ , in time δt seconds. Assuming the angular velocity ω to be constant, the angular momentum will now be represented by vector ox′.

Change in angular momentum.

$$= \vec{ox'} - \vec{ox} = \vec{xx'} = \vec{ox}.\delta\theta \qquad \text{...(in the direction of } \vec{xx'}\text{)}$$
$$= I.\ \omega.\delta\theta$$

and rate of change of angular momentum

$$= I.\omega \times \frac{\delta\theta}{dt}$$

Since the rate of change of angular momentum will result by the application of a couple to the disc, therefore the couple applied to the disc causing precession,

$$C = \underset{\delta t \to 0}{\mathrm{Lt}}\ I.\omega \times \frac{\delta\theta}{\delta t} = I.\omega \times \frac{d\theta}{dt} = I.\omega.\omega_P \qquad \ldots \left(\because \frac{d\theta}{dt} = \omega_P \right)$$

where ω_P = Angular velocity of precession of the axis of spin or the speed of rotation of the axis of spin about the axis of precession OY.

1.The couple I.ω.ωp, in the direction of the vector xx′ (representing the change in angular momentum) is the active gyroscopic couple, which must be applied over the disc when the axis of spin is made to rotate with angular velocity ω about the axis of precession. The vector xx′ lies in the plane XOZ or the horizontal plane. In case of a very small displacement δθ, the vector xx′ will be perpendicular to the vertical plane XOY. Therefore, the couple causing this change in the angular momentum will lie in the plane XOY. The vector xx′, as shown in Fig(b), represents an anticlockwise couple in the plane XOY. Therefore, the plane XOY is called the plane of active gyroscopic couple and the axis OZ perpendicular to the plane XOY, about which the couple acts, is called the axis of active gyroscopic couple.

2. When the axis of spin itself moves with angular velocity ω_P, the disc is subjected to reactive couple whose magnitude is same (i.e. $I.\omega.\omega_P$) but opposite in direction to that of active couple. This reactive couple to which the disc is subjected when the axis of spin rotates about the axis of precession is known as reactive gyroscopic couple. The axis of the reactive gyroscopic couple is represented by OZ′ in Fig(a).

3. The gyroscopic couple is usually applied through the bearings which support the shaft. The bearings will resist equal and opposite couple.

4. The gyroscopic principle is used in an instrument or toy known as gyroscope. The gyroscopes are installed in ships in order to minimize the rolling and pitching

effects of waves. They are also used in Aeroplanes, monorail cars, gyrocompasses etc.

Effect of the Gyroscopic Couple on an Aeroplane -

The top and front view of an aeroplane are shown in Fig (a). Let engine or propeller rotates in the clockwise direction when seen from the rear or tail end and the aeroplane takes a turn to the left.

Let ω= Angular velocity of the engine in rad/s,

m = Mass of the engine and the propeller in kg,

k = Its radius of gyration in meter

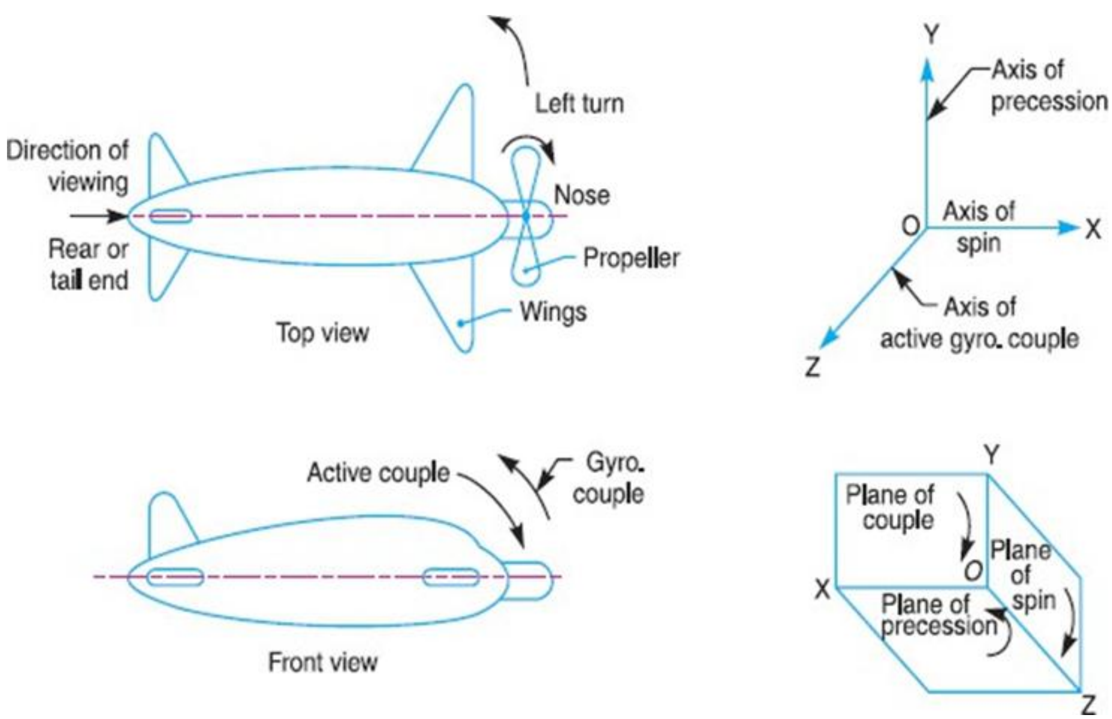
I = Mass moment of inertia of the engine and the propeller in kg-m^2

= $m.k^2$,

v = Linear velocity of the aeroplane in m/s,

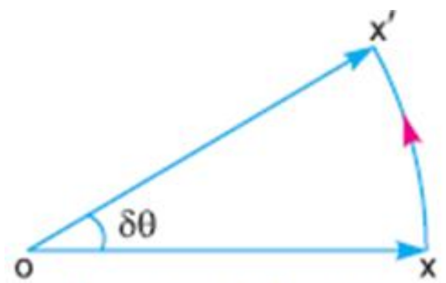
R = Radius of curvature in meters, and ω_P = Angular velocity of precession=V/R

Gyroscopic couple acting on the aeroplane, $C = I.\omega.\omega_P$

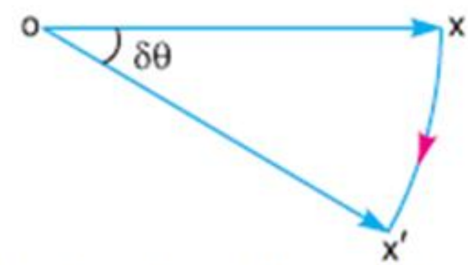


Before taking the left turn, the angular momentum vector is represented by ox. When it takes left turn, the active gyroscopic couple will change the direction of the angular momentum vector from ox to ox′ as shown in Fig(a). The vector xx′, in the limit, represents the change of angular momentum or the active gyroscopic couple and is perpendicular to ox. Thus, the plane of active gyroscopic couple XOY will be perpendicular to xx′, i.e. vertical in this case, as shown in Fig (b). By applying right hand screw rule to vector xx′, we find that the direction of active gyroscopic couple is clockwise as shown in the front view of Fig (a).

In other words, for left hand turning, the active gyroscopic couple on the aeroplane in the axis OZ will be clockwise as shown in Fig (b). The reactive gyroscopic couple (equal in magnitude of active gyroscopic couple) will act in the opposite direction (i.e. in the anticlockwise direction) and the effect of this couple is, therefore, to raise the nose and dip the tail of the aeroplane.



(*a*) Aeroplane taking left turn.



(*b*) Aeroplane taking right turn.

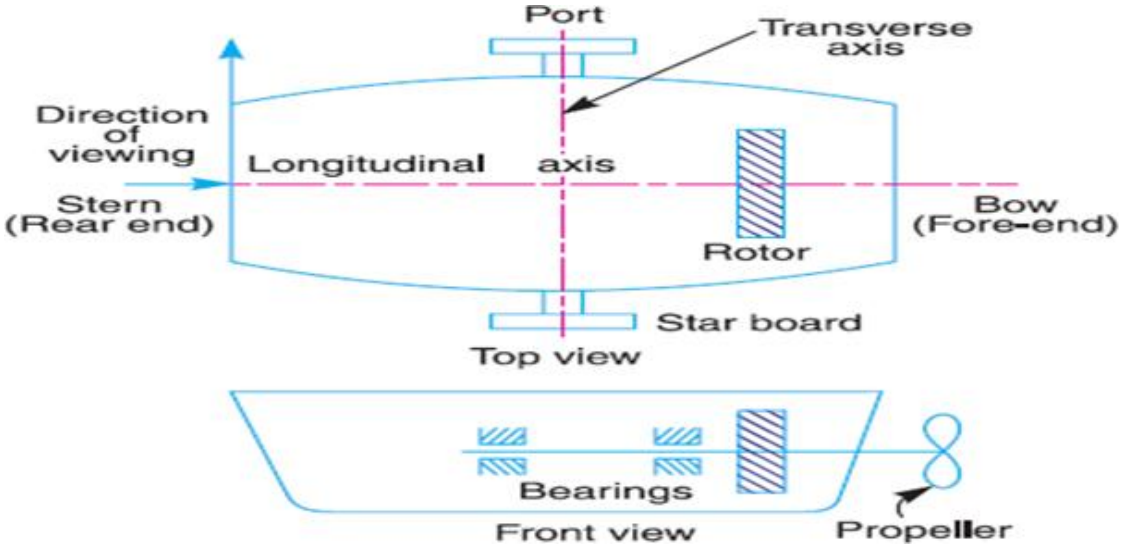
<u>Terms Used in a Naval Ship -</u>

The top and front views of a naval ship are shown in Fig 14.7. The fore end of the ship is called bow and the rear end is known as stern or aft. The left hand and right-hand sides of the ship, when viewed from the stern are called port and starboard respectively. We shall now discuss the effect of gyroscopic couple on the naval ship in the following three cases:

1. Steering,

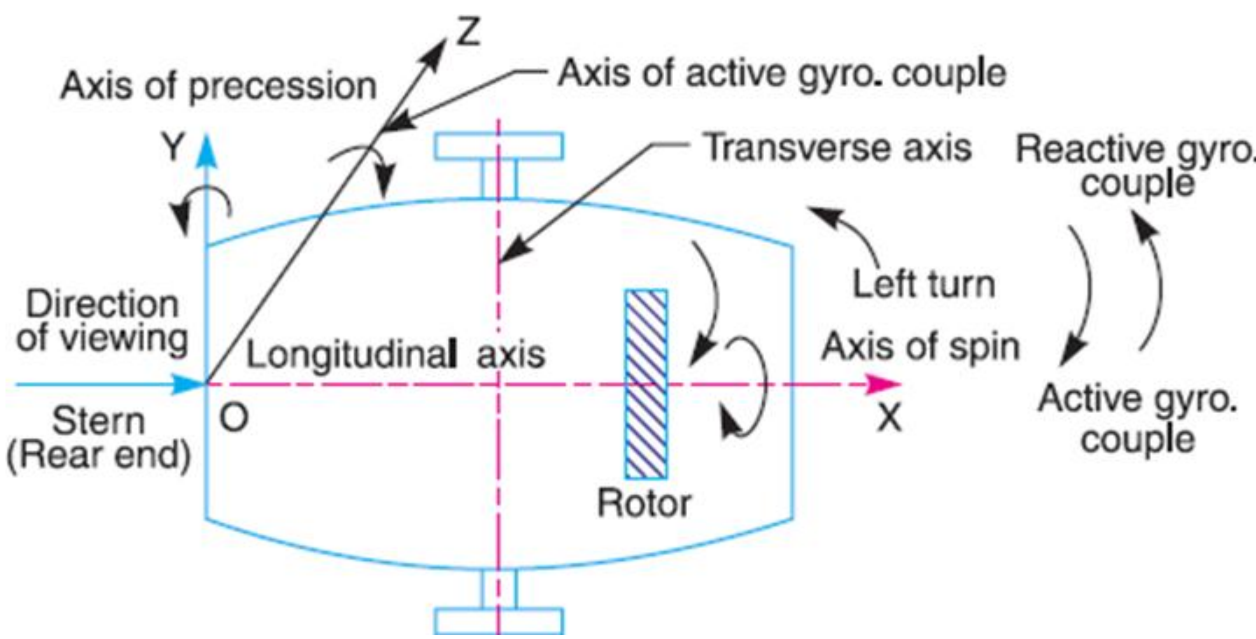
2. Pitching,

3. Rolling.

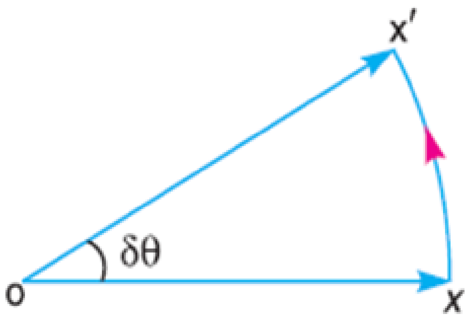


Effect of Gyroscopic Couple on a Naval Ship during Steering -

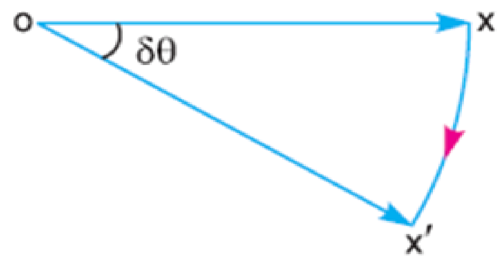
Steering is the turning of a complete ship in a curve towards left or right, while it moves forward. Consider the ship taking a left turn, and rotor rotates in the clockwise direction when viewed from the stern, as shown in Fig. The effect of gyroscopic couple on a naval ship during steering taking left or right turn may be obtained in the similar way as for an aeroplane.



When the rotor of the ship rotates in the clockwise direction when viewed from the stern, it will have its angular momentum vector in the direction ox as shown in Fig(a). As the ship steers to the left, the active gyroscopic couple will change the angular momentum vector from ox to ox′. The vector xx′ now represents the active gyroscopic couple and is perpendicular to ox. Thus, the plane of active gyroscopic couple is perpendicular to xx′ and its direction in the axis OZ for left hand turn is clockwise as shown in Fig. The reactive gyroscopic couple of the same magnitude will act in the opposite direction (i.e., in anticlockwise direction). The effect of this reactive gyroscopic couple is to raise the bow and lower the stern.



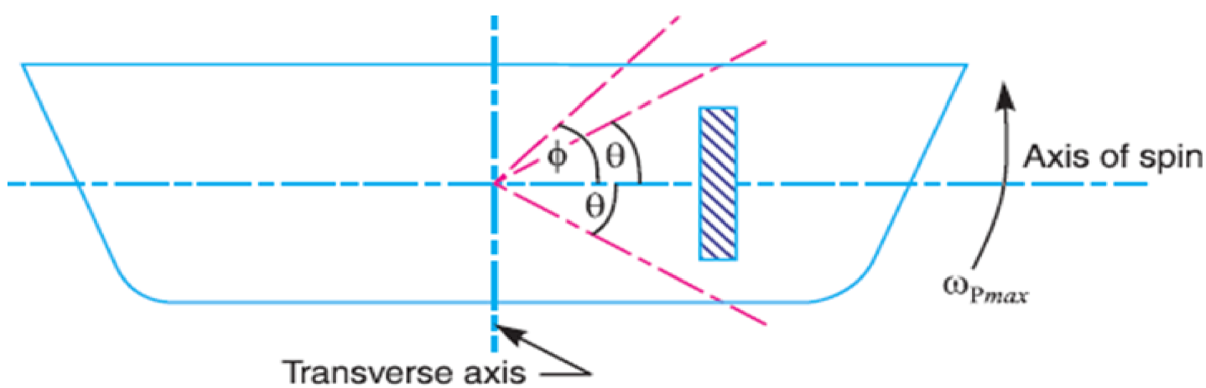
(a) Streeing to the left



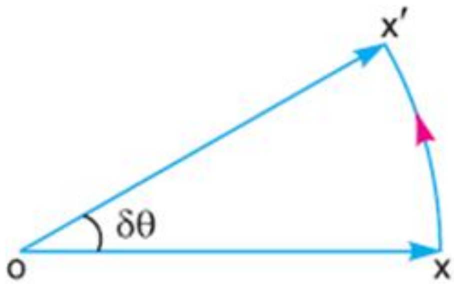
(b) Streeing to the right

Effect of Gyroscopic Couple on a Naval Ship during Pitching -

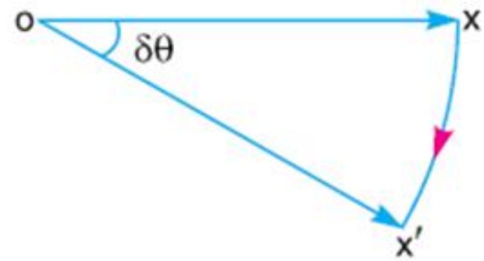
Pitching is the movement of a complete ship up and down in a vertical plane about transverse axis, as shown in Fig(a). In this case, the transverse axis is the axis of precession. The pitching of the ship is assumed to take place with simple harmonic motion i.e., the motion of the axis of spin about transverse axis is simple harmonic.



(a) Pitching of a naval ship



(*b*) Pitching upward



(*c*) Pitching downward

∴ Angular displacement of the axis of spin from mean position after time *t* seconds,

$$\theta = \phi \sin \omega_1 . t$$

ϕ = Amplitude of swing *i.e.* maximum angle turned from the mean position in radians, and

ω_1 = Angular velocity of S.H.M.

$$= \frac{2\pi}{\text{Time period of S.H.M. in seconds}} = \frac{2\pi}{t_p} \text{ rad/s}$$

Angular velocity of precession,

$$\omega_P = \frac{d\theta}{dt} = \frac{d}{dt}(\phi \sin \omega_1 . t) = \phi \omega_1 \cos \omega_1 t$$

The angular velocity of precession will be maximum, if $\cos \omega_1 . t = 1$.

∴ Maximum angular velocity of precession,

$$\omega_{Pmax} = \phi . \omega_1 = \phi \times 2\pi / t_p \qquad \text{...(Substituting } \cos \omega_1 . t = 1\text{)}$$

Let I = Moment of inertia of the rotor in kg-m2, and ω= Angular velocity of the rotor in rad/s.

Maximum gyroscopic couple,

$$C_{max} = I . \omega . \omega_{Pmax}$$

When the pitching is upward, the effect of the reactive gyroscopic couple, as shown in Fig.(b), will try to move the ship toward starboard. On the other hand, if the pitching is downward, the effect of the reactive gyroscopic couple, as shown in Fig(c), is to turn the ship towards port side.

Effect of Gyroscopic Couple on a Naval Ship during Rolling -

We know that, for the effect of gyroscopic couple to occur, the axis of precession should always be perpendicular to the axis of spin. If, however, the axis of precession becomes parallel to the axis of spin, there will be no effect of the gyroscopic couple acting on the body of the ship.

In case of rolling of a ship, the axis of precession (i.e., longitudinal axis) is always parallel to the axis of spin for all positions. Hence, there is no effect of the gyroscopic couple acting on the body of a ship.

Q1.

A uniform disc of diameter 300 mm and of mass 5 kg is mounted on one end of an arm of length 600 mm. The other end of the arm is free to rotate in a universal bearing. If the disc rotates about the arm with a speed of 300 r.p.m. clockwise, looking from the front, with what speed will it precess about the vertical axis?

Solution. Given: d = 300 mm or r = 150 mm = 0.15 m ; m = 5 kg ; l = 600 mm = 0.6 m ;N = 300 r.p.m. or $\omega = 2\pi \times 300/60 = 31.42$ rad/s

We know that the mass moment of inertia of the disc, about an axis through its centre of gravity and perpendicular to the plane of disc,

$I = m.r^{2/}2 = 5(0.15)2/2 = 0.056$ kg-m^2

couple due to mass of disc,

$C = m.g.l = 5 \times 9.81 \times 0.6 = 29.43$ N-m

Let ω_P = Speed of precession.

We know that couple (C),

$$29.43 = I.\omega.\omega_P = 0.056 \times 31.42 \times \omega_P = 1.76\ \omega_P$$

$$\omega_P = 29.43/1.76 = 16.7 \text{ rad/s}$$

Q2. An aeroplane makes a complete half circle of 50 meters radius, towards left, when flying at 200 km per hr. The rotary engine and the propeller of the plane has a mass of 400 kg and a radius of gyration of 0.3 m. The engine rotates at 2400 r.p.m. clockwise when viewed from the rear. Find the gyroscopic couple on the aircraft and state its effect on it.

Solution. Given: $R = 50$ m ; $v = 200$ km/hr = 55.6 m/s ; $m = 400$ kg ; $k = 0.3$ m ; $N = 2400$ r.p.m. or $\omega = 2\pi \times 2400/60 = 251$ rad/s

We know that mass moment of inertia of the engine and the propeller,

$I = m.k2 = 400(0.3)2 = 36$ kg-m^2

angular velocity of precession, $\omega P = v/R = 55.6/50 = 1.11$ rad/s

We know that gyroscopic couple acting on the aircraft,

$C = I.\ \omega.\ \omega P = 36 \times 251.4 \times 1.11 = 100\ 46$ N-m = 10.046 kN-m

when the aeroplane turns towards left, the effect of the gyroscopic couple is to lift the nose upwards and tail downwards.

Q3. The turbine rotor of a ship has a mass of 8 tonnes and a radius of gyration0.6m. It rotates at 1800 r.p.m. clockwise, when looking from the stern. Determine the gyroscopic couple, if the ship travels at 100 km/hr and steer to the left in a curve of 75 m radius.

Solution. Given: $m = 8$ t = 8000 kg ; $k = 0.6$ m ; $N = 1800$ r.p.m.

$\omega = 2\pi \times 1800/60 = 188.5$ rad/s ; $v = 100$ km/h = 27.8 m/s ; $R = 75$ m

We know that mass moment of inertia of the rotor,

$I = m.k^2 = 8000\ (0.6)2 = 2880$ kg-m^2

angular velocity of precession, $\omega P = v / R = 27.8 / 75 = 0.37$ rad/s

We know that gyroscopic couple,

$C = I.\omega.\omega P = 2880 \times 188.5 \times 0.37 = 200\ 866$ N-m

= 200.866 kN-m

when the rotor rotates in clockwise direction when looking from the stern and the ship steers to the left, the effect of the reactive gyroscopic couple is to raise the bow and lower the stern.